

Time-dependent reliability-based design optimization of main shaft bearings in wind turbines involving mixed-integer variables

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ABSTRACT

Main shaft bearings are essential transmission components in wind turbines. The failure of the main shaft bearings is inevitably accompanied by catastrophic consequences. In this regard, this paper presents a time-dependent reliability-based design optimization (TRBDO) approach for main shaft bearings involving mixed-integer variables. The maximization of fatigue life and the elastohydrodynamic film thickness under the load spectrum are selected as optimization objectives. The time-dependent reliability and geometric correlation are determined as nonlinear constraints. An efficient two-stage enrichment strategy is introduced to handle the time-dependent probabilistic constraint with mixed-integer design variables. A mixed-integer nonlinear optimization method is developed based on a meta-heuristic algorithm to solve the formulated TRBDO problem. Eventually, the effectiveness and robustness of the proposed approach are demonstrated by a real application in a 5 MW wind turbine.

1. Introduction

Renewable-based power generation technologies are essential for energy conservation and emission reduction in industries, among which wind power generation is one of the most sustainable and promising technologies [1]. According to the statistics from the Global Wind Energy Council [2], the cumulative installed wind turbine capacity grows rapidly, especially at off-shore locations with abundant wind energy resources. In addition, larger wind turbines are inevitable to increase the requirement of productivity and generation stability. These trends constitute challenges to the reliability of wind turbines. Therefore, a reasonable method for improving the reliability of the wind turbine life cycle in the design phase is urgently required.

Unscheduled repairs of critical components cause long-term downtimes for wind turbines and reduce the reliability and stability of power generation systems [3]. Numerous design approaches have been proposed for these components to enhance their comprehensive performance. Negm et al. [4] developed an optimization framework for wind turbine towers to achieve satisfactory dynamic characteristics. Li et al. [5] proposed an optimization method for wind turbine blades, which reasonably balances the strength and cost during the design process. Hegseth et al. [6] conducted an integrated optimization design of

platforms, tower structures, mooring systems, and pitch controllers in the wind turbine. Zhang et al. [7] utilized the finite element method (FEM) and topological optimization to obtain the optimal shape of the jacket structure in a wind turbine with appropriate mass. Lu et al. [8] established an optimization model for wind turbine tripod foundations to obtain satisfactory fatigue resistance with minimal volume. All the above investigations explored the failure mechanism of components and implemented diverse optimization designs from different perspectives. Nevertheless, inherent uncertainties such as manufacturing, assembly quality, and loads were not considered during the design process.

The reliability-based design optimization (RBDO) method has garnered widespread attention to address the uncertainties inherent during the design phase [9]. RBDO aims to balance cost and safety requirements harmoniously by integrating reliability as a functional constraint. In the past decade, researchers have proposed various strategies to address the RBDO problem, such as the two-level [10], mono-level [11], and decoupling approaches [12]. The two-level approach handles the RBDO problem by incorporating an inner loop for reliability assessment and an outer loop for optimization search. The mono-loop approach redefines the RBDO problem by substituting reliability constraints with Karush-Kuhn-Tucker optimality conditions [13]. The decoupling approach converts RBDO into deterministic

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optimizations by performing optimization and reliability analyses sequentially. Nevertheless, the aforementioned approaches are constrained by their respective implementation limitations. The primary challenges for mono-level and decoupling approaches are convergence and accuracy issues when dealing with multiple failure domains and highly nonlinear Limit State Function (LSF) [14], especially for high-fidelity numerical models of wind turbine components. Although the two-level approach demonstrates satisfactory accuracy, it suffers from a heavy computational burden due to the reliability assessment within the inner loop. To remedy this, some studies have replaced the expensive numerical model with a cheap surrogate model in the inner loop [15]. Benefiting from the development of surrogate model-based approaches, several RBDO researches have been conducted on critical components of wind turbines, including blades [16], towers [17], and floaters [18].

Compared to the RBDO, time-dependent reliability-based design optimization (TRBDO) incorporates time-dependent reliability as a constraint from the perspective of the entire lifecycle [19,20]. Thus, TRBDO can provide a relatively reasonable and reliable design for wind turbine components with long service life. In terms of this idea, Liu et al. [21] improved the fatigue reliability of gear transmission in wind turbines with suitable volume. All aforementioned investigations focused on probabilistic constraints with continuous variables. However, the design of the main shaft bearing involves a combination of both continuous variables and integer variables, such as the roller number. The additional integer design variables increase the complexity of handling probability constraints and conducting optimization procedures. The ordinary RBDO and TRBDO are inadequate to implement the design of the main bearing involving mixed-integer design variables. Hence, it is urgent to develop an efficient algorithm for this problem.

Although RBDO and TRBDO with mixed-integer variables have not been fundamentally conquered, several universal deterministic design approaches have already been developed for slewing bearings to mitigate this challenge [22,23]. These deterministic design techniques aim to enhance the bearing dynamic capacity, which directly impacts the fatigue life of bearings. However, the bearing dynamic capacity is independent of the loading conditions. Therefore, these approaches may result in unreasonable design and unnecessary expenses for different application scenarios. Additionally, internal contact characteristic analysis is unavoidable in the bearing design contexts to account for the randomness of loads, structure, and assembly quality. Nevertheless, the complex bearing structural configurations and loading conditions result in highly nonlinear mechanical models, which bring convergence and efficiency issues [24]. These issues are particularly pronounced in large main shaft bearings with massive internal contact regions.

According to the literature reviewed above, it can be concluded that numerous nonlinear constraints, mixed-integer design variables, and model convergence issues constitute obstacles to the implementation of RBDO and TRBDO in the main shaft bearing. Motivated by the above analyses, a TRBDO approach for the main shaft bearings involving mixed-integer design variables is developed in this paper. The main innovative aspects or potential contributions are as follows:

- 1) A TRBDO framework that simultaneously considers multiple-axis stochastic loads, mixed-integer design variables, and nonlinear constraints is proposed.
- 2) An efficient and robust main shaft bearing model suitable for the proposed TRBDO under multi-axis random loads and diverse random structures is applied.
- 3) A practical TRBDO case for main shaft bearings involving mixed-integer variables is provided.

The remainder of this paper is organized as follows. In Section 2, the mathematical model of the TRBDO for the main shaft bearing is established with the multiple-axis stochastic loads, mixed-integer design variables, and nonlinear constraints. In Section 3, a two-stage

enrichment strategy is developed for time-dependent probabilistic constraints with mixed-integer design variables. Moreover, the mixed-integer nonlinear optimization (MINO) approach based on a meta-heuristic algorithm is developed for solving the formulated TRBDO problem. In Section 4, the robustness and effectiveness of the proposed TRBDO method are investigated. Finally, the paper is summarized, and the main conclusions are drawn in Section 5.

2. The proposed TRBDO model

The main shaft bearings are inevitably subject to fatigue deterioration during the long-term operation of wind turbines [25]. The degree of degradation will accumulate over time and further impact the reliability of wind turbines. To obtain reliable structural configurations of main shaft bearings, this section presents a TRBDO problem that considers multiple-axis random loads, mixed-integer design variables, and various nonlinear constraints.

The mathematical model of the proposed TRBDO problem with mixed-integer variables can be expressed as:

$$\begin{aligned} \text{find : } \mathbf{X}_d &= [x_{c1}, x_{c2}, \dots, x_{ck}, x_{d1}, x_{d2}, \dots, x_{dp}] \\ \min \mathbf{F} &= [f_1(\mathbf{X}_d), f_2(\mathbf{X}_d), \dots, f_i(\mathbf{X}_d)] \\ \text{s.t. } &\begin{cases} p_j(\mathbf{X}, t) \leq 0, j = 1, \dots, m \\ h_l(\mathbf{X}_d) \leq 0, l = 1, \dots, n \\ x_{dp} \in I_p^n, x_{ck} \in R_k^n \\ \mathbf{X}_d^L \leq \mathbf{X}_d \leq \mathbf{X}_d^U \end{cases} \end{aligned} \quad (1)$$

where $\mathbf{X}_d$ is the design variable vector, t is the designed lifespan of interest, x_{ck} is the k^{th} continuous variable, x_{dp} is the p^{th} integer variable, $f_i(\cdot)$ is the i^{th} objective function, $I_p^n \subseteq \{1, 2, \dots, n\}$ is the set of the p^{th} integer variable, R_k^n is the set of the k^{th} continuous variable, $p_j(\cdot)$ and $h_l(\cdot)$ are the time-dependent probability and geometric constraints, respectively, $\mathbf{X}_d^L$ and $\mathbf{X}_d^U$ are the lower and upper boundary vectors for $\mathbf{X}_d$, respectively.

2.1. Mixed-integer variables of the main shaft bearing

Tapered roller bearing (TRB) is one of the most typical rotating support structures for the main shaft of wind turbines [26]. TRB comprises the inner and outer raceways, rollers, and a cage, as shown in Fig. 1 (a). The geometric dimension details of the TRB are illustrated in Figs. 1 (b) - (c). In the global radial sectional view, the d_m is the pitch diameter, d is the bore diameter, D is the outer diameter, φ_j is the azimuth angle of the j^{th} roller, z is the number of rollers, and θ_c is the angular gap of the cage. In the local axial cross-sectional view, C and B are the widths of the outer and inner raceways. O_r is the rear face thickness of the outer raceway, d_{rl} is the rear flange diameter, I_l is the width of the flange, α_f is the flange angle, and d_{fr} is the front flange diameter.

A magnified view of the roller is depicted in Fig. 2. α_e and α_i are the contact angles of the outer and inner raceways, α_m is the mean contact angle, ε is the Semi-taper angle, D_w and d_w are the roller large-end and small-end diameters, d_e is the mean roller diameter, L_w is the contact length of the roller, L_r and L_a are the radial and axial assembly clearances. The main shaft bearings of wind turbines generally adopt logarithmic modification processing to reduce stress concentration at the roller edges. The roller profile function $h(\cdot)$ can be expressed as:

$$h(x) = K_x d_e \ln \left(1 / \left(1 - (2x/L_w)^2 \right) \right), \quad (2)$$

where K_x is the coefficient affecting the profile convexity.

The operational performance of the main shaft bearings is defined by the internal geometric shape. Not altering the transitional connection between the bearing and external components (bearing pedestal and main shaft), the geometric parameters such as the mean roller diameter

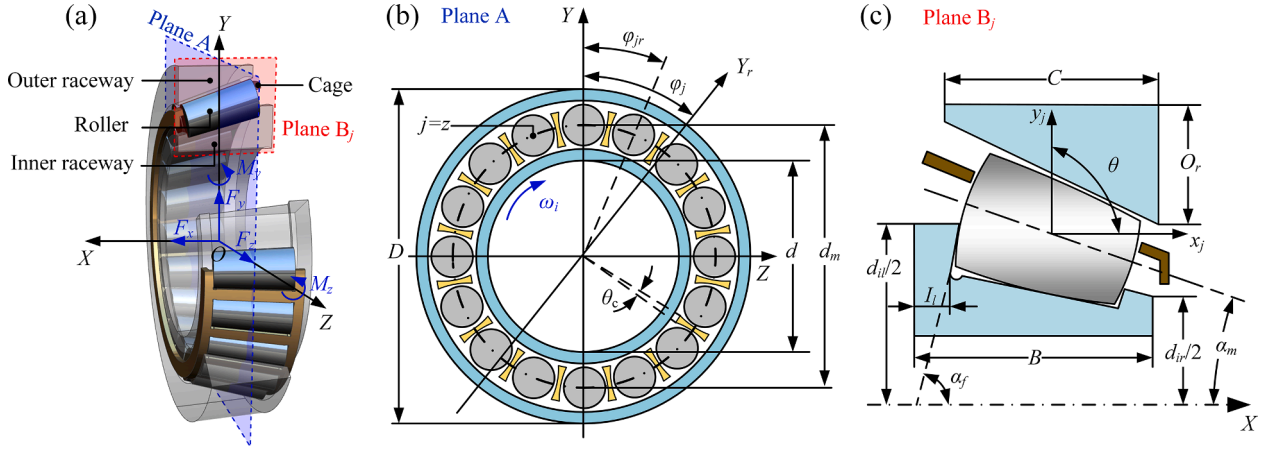


Fig. 1. Schematic diagram of a TRB geometry: (a) the mechanical model of TRB in 3-D view, (b) the global sectional view of TRB, (c) the local axial cross-sectional view of TRB in plane B_j ($j = 1 - z$).

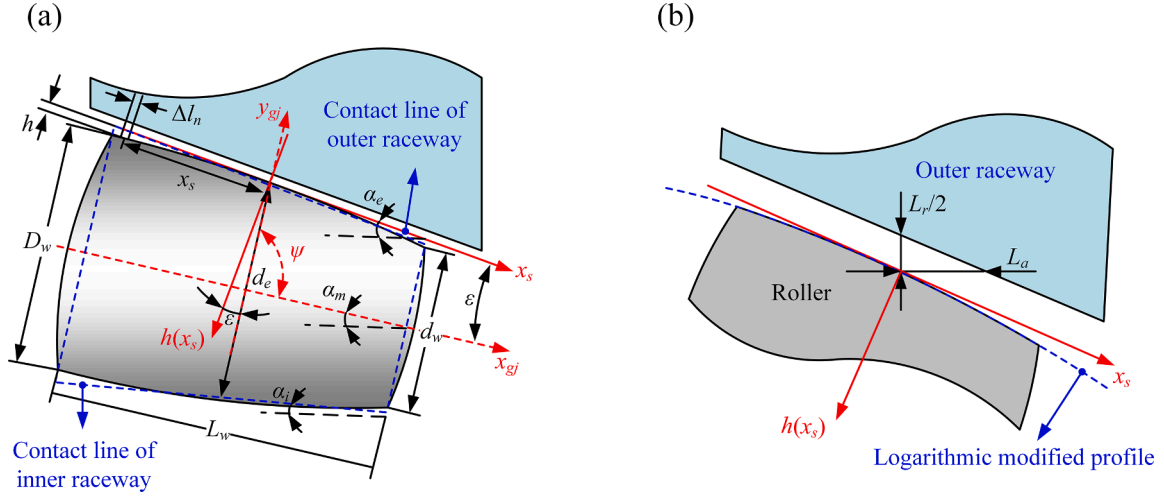


Fig. 2. Schematic diagram of roller geometry: (a) partially enlarged view of the roller, (b) partially enlarged view between the roller and outer raceway.

d_e , the contact length of the roller L_w , contact angles of the outer and inner raceways α_e and α_i , the pitch diameter d_m , the flange angle α_f , and roller profile coefficient K_y are selected as continuous design variables. The roller number z is selected as an integer design variable.

2.2. Objective function

The primary mission of the proposed TRBDO is improving the fatigue life and lubrication status of the main shaft bearing; the former supports service life, while the latter provides wear and micro-crack resistance for main shaft bearings. In real-life scenarios, wind turbines are subject to variable combined loads due to constantly changing weather conditions [27]. The statistical characteristics of the stochastic loads should be determined first to reasonably assess the fatigue life and lubrication status of the main shaft bearing, as shown in Fig. 3. The rain-flow counting method is adopted to count load duration distributions (LDD) of TRB obtained by force transducers. The fatigue load spectrum is transformed by compressing LDD [28], as shown in Fig. 3 (c).

Combined with the fatigue load spectrum and linear Palmgren-Miner theory [29], the fatigue life L of the TRB under stochastic loads can be derived as:

$$L = 1 / \sum_{c=1}^m (t_c / L_c) \times 10^6 \text{ rev}, \quad (3)$$

where

$$L_c = \left[(Q_{ci}/Q_{ei})^{-9/2} + (Q_{ce}/Q_{eo})^{-9/2} \right]^{-8/9} \times 10^6 \text{ rev}, \quad (4)$$

in which t_c and L_c are the time ratio and basic fatigue life of the c^{th} load case in the fatigue load spectrum, respectively, Q_{ci} and Q_{ce} are the dynamic capacities of the inner and outer raceway, which can be obtained by Appendix B. Parameters Q_{ei} and Q_{eo} are the equivalent dynamic loads of nonrotating and rotating raceway under the c^{th} load case. The values of Q_{ei} and Q_{eo} can be obtained by solving the mechanical model of TRB, which is provided in Appendix A. Therefore, the objective function f_1 can be expressed as:

$$f_1(X_d) = 1/L(X_d), \quad (5)$$

Sufficient lubrication oil separates the roller and raceway by forming an oil film, reducing friction, and enhancing contact efficiency. In fluid mechanics, the lubrication status between the roller and the raceway is considered elastohydrodynamic lubrication (EHL). Based on the theory of Hamrock and Dowson [30], the minimum oil film thickness $h_{\min}^0$ under fatigue load spectrum can be expressed as:

$$h_{\min}^0 = \min_{\substack{j \in \{1, 2, \dots, z\} \\ c \in \{1, 2, \dots, 16\}}} \left\{ \frac{2.65 L_w^{0.13} a_p^{0.54} \eta_o^{0.7} U_{e/i}^{0.7} R_{e/i}^{0.43}}{2^{0.7} Q_{ce/j/cij}^{0.13} \bar{E}^{0.03}} \right\}, \quad (6)$$

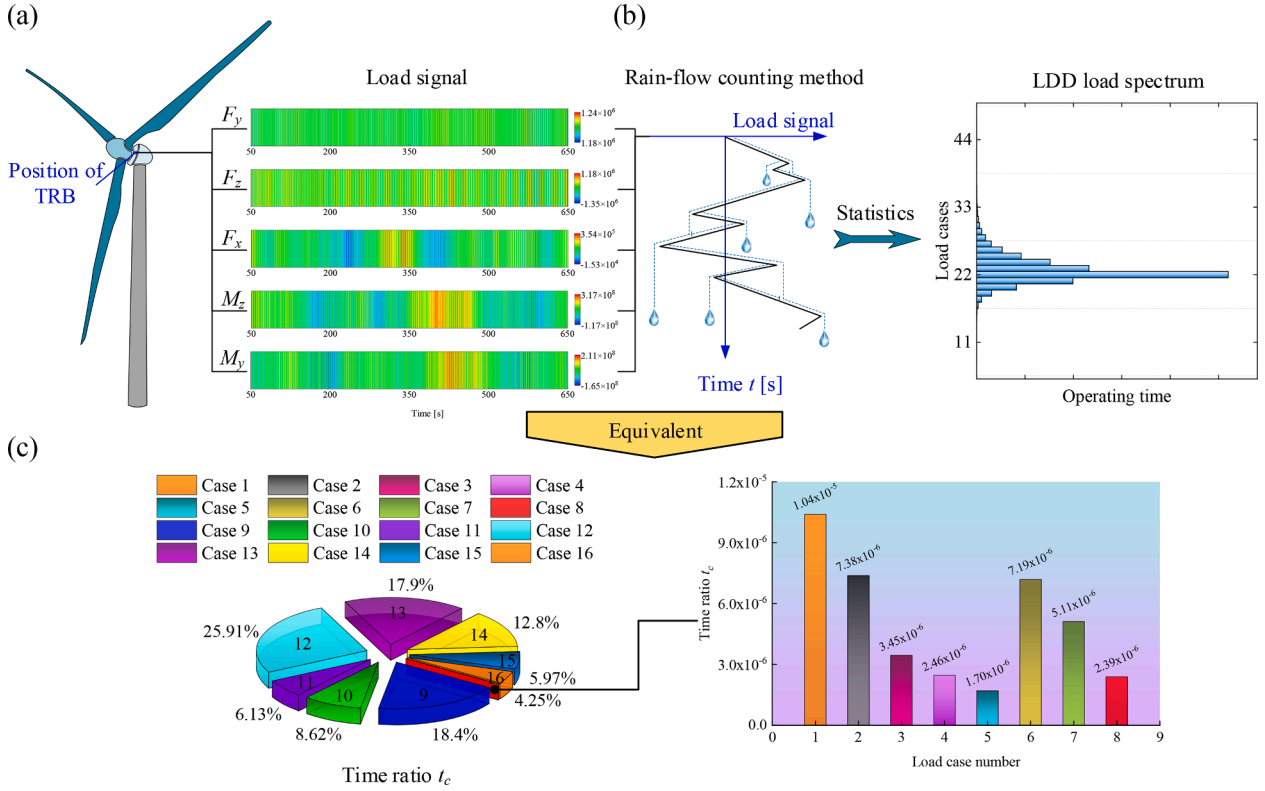


Fig. 3. Schematic diagram of fatigue load spectrum compilation: (a) collection of load signal, (b) statistics of the load duration distribution (LDD), and (c) equivalent of fatigue load spectrum.

where η_0 is the dynamic viscosity of the lubricant, a_p is the viscosity pressure coefficient, U_i and U_e are the entrainment velocities for the roller and raceways, respectively, the Q_{cej} and Q_{cij} are the contact loads of the j^{th} roller under the c^{th} load case and they can be obtained by fatigue load spectrum and **Appendix A**, $\bar{E}$ is the equivalent elastic modulus, $R_{e/i}$ is the curvature radius. Therefore, the objective function f_2 can be expressed as:

$$f_2(\mathbf{X}_d) = 1/h_{\min}^0(\mathbf{X}_d). \quad (7)$$

The proposed optimization model involves multiple interrelated objectives, which is referred to as multi-objective optimization. The weight vector-based and Pareto set-based methods are frequently employed to handle such problems [31]. Although the Pareto set-based method can provide decision-makers with more non-dominated design points in the Pareto Front, its convergence speed is considerably slower than the weight vector-based method. Thus, the weight vector-based method is adopted in this paper to handle the proposed TRBDO.

According to the duality principle, each objective f_i is independently optimized to obtain the unreachable maximum and minimum sets $\mathbf{F}_{\max}^* = [f_{1\max}^*, f_{2\max}^*, \dots, f_{n\max}^*]$ and $\mathbf{F}_{\min}^* = [f_{1\min}^*, f_{2\min}^*, \dots, f_{n\min}^*]$. The potential optimization weight ω_i for each objective can be defined as:

$$w_i(\mathbf{X}_d) = (f_i(\mathbf{X}_d) - f_{i\min}^*) / (f_{i\max}^* - f_{i\min}^*), \quad (8)$$

Subsequently, the multi-objective f_1 and f_2 are normalized as a single-objective f_{obj} , which can be expressed as:

$$f_{obj}(\mathbf{X}_d) = \sum_{i=1}^n \omega_i(\mathbf{X}_d). \quad (9)$$

2.3. Geometric constraints

The appropriate constraints can be utilized to refine the initial

sample space to a feasible design space. In addition, an unreasonable geometric structure of the bearing may lead to non-convergence of the mechanical model. Thus, the detailed geometric constraints are derived in this subsection.

Angular clearance of each roller should exist to avoid any collision and ensure the safe operation of TRB. This angular clearance affects the maximum roller number. Therefore, the constraint h_1 can be expressed as:

$$h_1(\mathbf{X}_d) = z\{2\arcsin[d_e \cos(\alpha_m)/d_m] + \theta_c\} - 360, \quad (10)$$

where z is the roller number and $\theta_c = 0.5^\circ$ is the angular gap of the cage.

The minimum number of rollers is restricted by the maximum size of the mean roller diameter, and it can be expressed as:

$$h_2(\mathbf{X}_d) = \frac{2\pi d_m}{D-d} - z \quad (11)$$

The mean roller diameter d_e is restricted by the thickness of the local axial cross-sectional (Fig. 1 (c)). Therefore, the constraint h_3 can be expressed as:

$$h_3(\mathbf{X}_d) = d_e - \frac{(D-d)}{2\cos(\alpha_m)}. \quad (12)$$

The flange contact points of the rollers should be limited by the mean roller diameter during the design of the inner ring flange:

$$h_4(\mathbf{X}_d) = (B-C) - \left(I_l + \frac{d_e \sin(\alpha_f)}{2}\right), \quad (13)$$

where I_l is the minimum flange width.

The contact length L_w of the roller should be restricted by the widths of the outer raceway and inner raceway, which can be written as:

$$\begin{cases} h_5(\mathbf{X}_d) = L_w - C/\cos(\alpha_e) \\ h_6(\mathbf{X}_d) = L_w - (B-I_l)/\cos(\alpha_i) \end{cases}, \quad (14)$$

where C and B are the widths of the outer and inner raceways, respectively.

The position of the roller must never interfere with the groove of the inner raceway. Therefore, the constraint h_7 can be expressed as:

$$h_7(\mathbf{X}_d) = L_w \sin(\alpha_i) - \frac{d_{il} + d_{ir} - 2d}{2}, \quad (15)$$

where d_{ir} and d_{il} are the front and the rear diameters of the flange, respectively.

According to the geometry characteristic of TRB, the contact angles of the outer raceways must be larger than those of the inner raceway. Therefore, the semi-taper angle should be ensured by:

$$h_8(\mathbf{X}_d) = \frac{(a_e - a_i)}{2} - \theta_{elim}, \quad (16)$$

where $\theta_{elim} = 2^\circ$ is the maximum angle of the semi-taper angle for the main shaft bearing.

The contact position between the roller large-end and flange must be less than the line of the mean contact angle, as shown in Fig. 1 (c). Therefore, the constraint h_9 can be expressed as:

$$h_9(\mathbf{X}_d) = 90 - a_m - a_f. \quad (17)$$

The convexity of the profile degree for the roller should be limited by the mean roller diameter:

$$h_{10}(\mathbf{X}_d) = K_s d_e \ln \left(1 / \left(1 - ((L_w - 2c_{1/2}) / L_w)^2 \right) \right) - d_e / 2, \quad (18)$$

where c_1 and c_2 are the corner radius of the roller large-end and small-end, respectively.

In engineering applications, the pitch diameter should be limited to ensure the stability of TRB operation [22]. Thus, these constraints can be expressed as:

$$\begin{cases} h_{11}(\mathbf{X}_d) = (0.5 - e_{\min})(D + d) - d_m \\ h_{12}(\mathbf{X}_d) = d_m - (0.5 + e_{\max})(D + d) \end{cases}, \quad (19)$$

where $e = (0.01-0.07)$ is a range coefficient for TRB design, d is the bore diameter, and D is the outer diameter.

The rear face of the outer raceway should be restricted by the width and contact angle of the outer raceway:

$$h_{13}(\mathbf{X}_d) = C \tan(\alpha_e) - O_r, \quad (20)$$

where O_r is the rear face thickness of the outer raceway.

2.4. Time-dependent probability constraint

Most of the previous studies are carried out under deterministic assumptions. However, in practice, the parameters of wind turbines are uncertain due to variations in manufacturing, assembly, and operation processes. Hence, the time-dependent reliability theory is adopted to handle the uncertainties inherent in the main shaft bearing.

The time-dependent failure probability of the TRB can be expressed as:

$$P_F(0, T) = \mathbf{prob} \left\{ g \left(\underbrace{\mathbf{X}_d}_{\in \mathbf{X}_d}, \underbrace{x_1, x_2, t}_{\in \mathbf{X}_R} \right) \leq 0, \exists t \in [0, T] \right\}, \quad (21)$$

where the time-dependent LSF is

$$g \left(\underbrace{\mathbf{X}_d}_{\in \mathbf{X}_d}, \underbrace{x_1, x_2, t}_{\in \mathbf{X}_R} \right) = L(x_{d1}, \mathbf{X}_R) - C_t N_e t, \quad (22)$$

in which $\mathbf{X}_d$ is the design variable vector, $\mathbf{X}_R = [d_e, L_w, \alpha_e, \alpha_i, d_m, \alpha_f, K_s, L_r,$

$N_e]^T$ is the random variable vector for TRB, $C_t = 365 \times 24 \times 60$ is the coefficient of unit conversion, L_r is the radial assembly clearance, N_e is the revolutions per minute (rpm) of TRB, t is the design lifespan, and L is the fatigue life of the TRB under stochastic loads. It should be noted that x_{d1} is an integer variable representing roller number z of the main shaft bearing. Thus, the failure and safe region of time-dependent LSF in the observation period can be defined as:

$$\begin{cases} F = g(z, \mathbf{X}_R, t) \leq 0, \exists t \in [0, T] \\ S = g(z, \mathbf{X}_R, t) > 0, \forall t \in [0, T] \end{cases}. \quad (23)$$

The time-dependent probability constraint can be expressed as:

$$p_1(\mathbf{X}_d) = R_t^T - (1 - P_F(z, \mathbf{X}_R, t)), \quad (24)$$

where R_t^T is the target reliability. For the TRBDO of the main shaft bearing, the constraint sets are normalized with the objective function to apply as a penalty function:

$$f_R = f_{obj}(\mathbf{X}_d) + \sum_{l=1}^n k_{dl} |h_l(\mathbf{X}_d)| + \sum_{j=1}^m k_{pj} |p_j(\mathbf{X}_d)|, \quad (25)$$

where k_{dl} and k_{pj} are the penalty factors of the l^{th} geometric constraint and the j^{th} time-dependent probability constraint. Although h_l and p_j are mandatory constraints of the TRBDO, their attainment difficulty levels are different. Suitable penalty factors should be selected to avoid violations of these constraints. The maximum value of the normalized geometric constraint functions $h^{\text{sum}}_{dl}(\cdot)$ can be determined based on the boundary values of design variables. Besides, the range of reliability constraint function ranges from $R_t^T - 1$ to R_t^T . Therefore, the penalty factors k_{dl} and k_{pj} can be obtained by:

$$k_{dl} = \begin{cases} 0 & \text{if } h_l(\mathbf{X}_d) \leq 0 \\ \Delta_d & \text{otherwise} \end{cases}, k_{pj} = \begin{cases} 0 & \text{if } R_t^T - 1 \leq p_j \leq 0 \\ h^{\text{sum}}_{dl}(\mathbf{X}_d) \Delta_d / R_t^T & \text{otherwise} \end{cases}, \quad (26)$$

where $\Delta_d = 10^6$ is the penalty baseline. It should be noted that the convergence of the bearing mechanical model may be compromised for unfeasible geometric configurations. Hence, k_{dl} must be evaluated before computing the penalty function value f_R . If k_{dl} is non-zero, evaluating the objective function f_{obj} and reliability constraints p_j becomes futile. In this scenario, the algorithm sets f_{obj} as a maximum value of 2 and P_F as 100%.

In addition, the time-dependent probability constraint defined by Eq. (24) cannot be directly solved by analytical and numerical methods due to the high dimensionality and integration complexity [20]. The Monte Carlo Simulation (MCS) method is an appropriate option for the proposed probability constraint. Although MCS is relatively easy to apply, its accuracy is dominated by the large-scale evaluation of time-dependent LSF, which causes a heavy computational burden [15]. Thus, an efficient two-stage enrichment-based TRBDO approach is developed in the following section to handle time-dependent probabilistic constraints with mixed-integer variables.

3. Solution of the TRBDO model

The primary emphasis of the proposed TRBDO is effectively handling the time-dependent probabilistic constraint with integer design variables. Therefore, two approximation strategies named global and local enrichments are proposed in this section. Furthermore, a metaheuristic MINO approach is developed to implement the proposed TRBDO.

3.1. Global enrichment of time-dependent LSF

The application of surrogate models can improve the efficiency of TRBDO. Among various surrogate models, the Kriging model is widely applied in TRBDO due to its rational adaptive learning and uncertainty

quantification capabilities [32]. Conventional Kriging-based TRBDO necessitates approximating time-dependent LSFs for each design variable during the optimal exploration. However, the stochastic discretization and corresponding time-dependent LSF approximation for each design variable are ponderous and time-consuming. An ideal approach is to proactively anticipate the limit state information of the entire design space before the optimization process [33]. As long as the limit state information is sufficiently acquired, the probability constraint assessment will become efficient and precise. For the proposed TRBDO of the main shaft bearing, the additional integer design variable complicates the acquisition of limit state information. Therefore, a unique Kriging model involving mixed-integer variables is preconstructed in this subsection to obtain limit state information within the design space.

According to Eq. (1), the probability density function (PDF) of random variables varies within the design space. The statistical information of random variables in the design space can be defined as an augmented PDF, which covers artificial (for design variables) and aleatory uncertainties (for random variables) simultaneously [14]. Fig. 4 illustrates the difference between the original and the augmented PDF. The augmented PDF $h_X(x)$ can be expressed as:

$$h_X(x) = \int_{D_p} f_X(x|\mu)\pi(\mu)d\mu. \quad (27)$$

The feasible design space for the formulated TRBDO with mixed-integer variables must be reasonably defined to achieve global enrichment of the time-dependent LSF. The 3-dimensional TRBDO with three types of variables is taken as an example, as shown in Fig. 5. The blue ball represents the non-designable random variable R. Green and yellow balls represent the integer design variable DD and designable random variable DR, respectively. Parameters DD_i and DR_i are generated by discretized integer and augmented PDF, respectively. A combination strategy is performed in Fig. 5 (a). It should be noted that the LSF is generally composed of high-fidelity models such as finite element models and nonlinear numerical models. Unreasonable design variables can lead to non-convergence or singular solutions in the LSF. Hence, the feasible space must be redefined, as shown in Fig. 5 (b). This step primarily involves implementing the geometric constraints established in Section 2.3. Subsequently, the global enrichment can be conducted in terms of the determined feasible space. The adaptive Kriging-based global enrichment strategy is presented as follows:

Step 1. The N^x_G -size trajectory matrix $S^x_G = [X_{1G}, X_{2G}, X_{3G}, \dots, X_{N^x_G}]$ is generated according to Fig. 5. The N^t -size time vector $S^t_G = [t_1, t_2, t_3, \dots, t_{N^t}]$ for the observation region is uniformly dispersed. The N^j -size initial training trajectories (X^T_j, t^T_j) ($j=1, \dots, N^j$) are randomly extracted from S^x_G and S^t_G .

Step 2. The initial training set $T_G = [X^T_j, t^T_j, g(X^T_j, t^T_j)]$ is generated by N^j repeated evaluation of the proposed time-dependent LSF. Subsequently, the initial Kriging surrogate model $g_{kG}(X, t)$ is established by T_G .

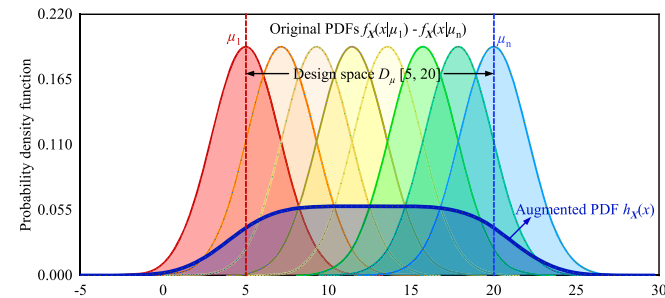


Fig. 4. Schematic diagram of predefined augmented probability density function.

The linear regression and Gaussian correlation functions are selected to establish the global Kriging surrogate model.

Step 3. The U -function [34] is adopted to quantify the approximation level of the Kriging model to the time-dependent LSF in the feasible space:

$$U_{\min}(X_i) = \frac{|\mu_{gk}(X_i, t_j)|}{\sigma_{gk}(X_i, t_j)} \quad j = 1, 2, \dots, N^t, \quad (28)$$

where $\mu_{gk}(X_i, t_j)$ and $\sigma_{gk}(X_i, t_j)$ are the predicted value and corresponding variance of the Kriging surrogate model. These values can be calculated via Appendix C. The new training trajectory X^{up}_{iG} for the global feasible design space is determined as:

$$X^{up}_{iG} = \arg \min_{X_{iG} \in S^x_G} U_{\min}(X_{iG}). \quad (29)$$

The corresponding update time variable t^{up}_{jG} is determined as:

$$t^{up}_{jG}(X^{up}_{iG}) = \arg \min_{t_j \in S^t_G} \{U_{\min}(X^{up}_{iG}, t_j)\}. \quad (30)$$

Step 4. According to [34], the accuracy of Kriging for each time trajectory X_i within feasible space is represented by the probability $\Phi(U_{\min}(X_i))$. If $U_{\min}(X_i) \geq 2$, the probability of correctly identifying the sign for the time-dependent LSF is greater than or equal to $\Phi(2) = 97.7\%$. Thus, the learning process is terminated by $U_{\min}(X_i) \geq 2$. Otherwise, go to Step. 2 with the new training set $T = [T \cup (X^{up}_{iG}, t^{up}_{jG}, g(X^{up}_{iG}, t^{up}_{jG}))]$.

Step 5. The training set T_G -based Kriging model $g_{kG}(X, t)$ is recorded and the global enrichment process is terminated.

3.2. Local enrichment of probabilistic constraint

Conducting global enrichment of the time-dependent LSF obtains crucial limit state information within the augmented and time domains. Distinct from global enrichment, local enrichment aims to accurately evaluate the probability constraints during the optimization exploration phase. Hence, it is crucial to transition primary attention from the global feasible domain to the local design point.

For this purpose, the accuracy of the obtained Kriging model $g_{kG}(X, t)$ should be evaluated for the local limit state during optimization. If needed, further learning should be conducted to enhance its accuracy. In addition, the integer design variables are fixed values after specifying the design variables in each optimization iteration. Correspondingly, the trajectory matrix of local enrichment S^x_L can be formed by the joint PDF and fixed value of the specific integer.

According to [32], the single-loop Kriging (SILK) is regarded as an effective approach for solving time-dependent probabilities. Therefore, SILK is introduced in local enrichment for evaluating the time-dependent failure probability. Similar to the global enrichment, the local enrichment also utilizes linear regression and Gaussian correlation functions to construct the Kriging model. The stop criterion for the SILK method can be expressed as follows:

$$\varepsilon_{\max} = \max_{N_{fb} \in (0, N^x_2)} \left\{ \frac{|N_{fb} - N^x_{fb}|}{N_{fa} + N^x_{fb}} \right\} < \varepsilon_{\text{elim}}, \quad (31)$$

where $\varepsilon_{\text{elim}} = 0.05$ is the threshold value for the stop criterion, N_{fa} and N_{fb} are precise and imprecise time trajectories in S^x_L . For simplicity, the specific details of the SILK can be referred to [35]. The time-dependent failure probability of the main shaft bearing can be estimated as:

$$\hat{P}_F(0, T) = (N_{fa} + N_{fb}) / N^x_{MCS}, \quad (32)$$

where N^x_{MCS} is the time trajectory scale of S^x_L . Furthermore, a quantitative evaluation of the time trajectory matrix S^x_L is required to ensure

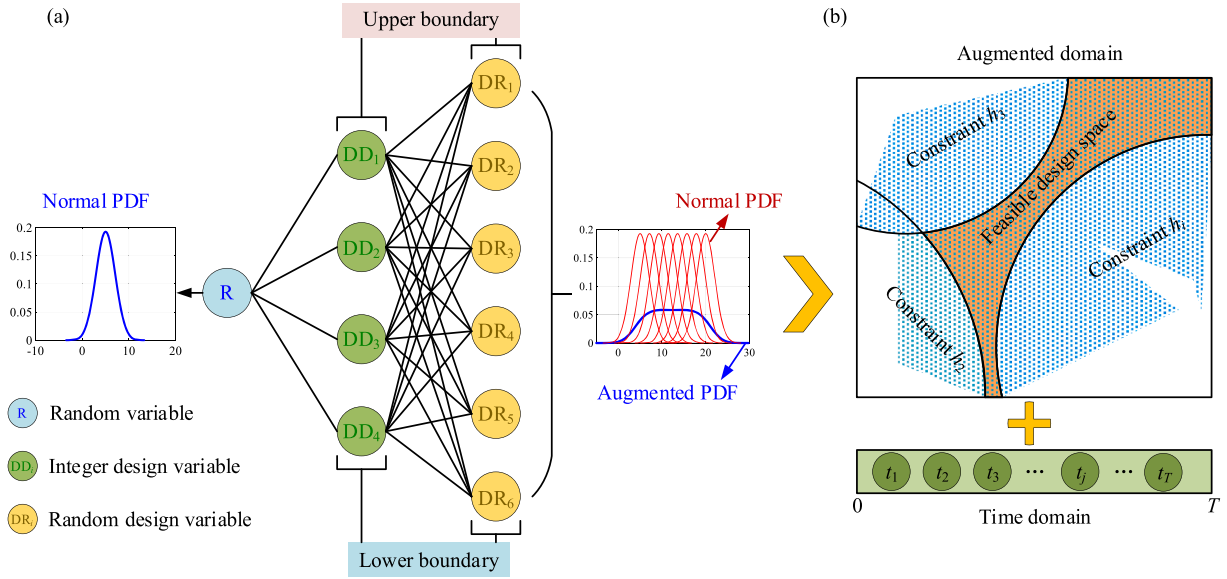


Fig. 5. Schematic diagram of predefined feasible space for global enrichment: (a) the combination of three types of variables, (b) generation of time domain and feasible space.

the unbiasedness of time-dependent reliability results. In this regard, the coefficient of variation (COV) of time-dependent failure probability is employed to validate the sufficiency of the trajectory matrix:

$$\text{Cov}(\hat{P}_F) = \sqrt{(1 - \hat{P}_F) / [\hat{P}_F(N_{MCS}^x - 1)]} < \text{Cov}_{\text{lim}}, \quad (33)$$

where $\text{Cov}_{\text{lim}}=5\%$ is the threshold value of Cov. If $\text{Cov}(\hat{P}_F) < 5\%$, the failure probability evaluation is considered acceptable. Otherwise, the trajectory matrix $\mathbf{S}_L^x$ should be expended with $\Delta N_{SL} = 10^4$ time trajectories [36].

Once the reliability constraint is determined, the TRBDO problem can be solved by diversified nature-inspired optimization algorithms. These methods can handle various linear or nonlinear optimization problems in a multidimensional continuous design space. Nevertheless, for the proposed TRBDO, the design variable, such as the roller number, belongs to the integer design space. Therefore, a MINO method based on

Hyper-Spherical Search (HSS) is developed in the following subsection to implement proposed TRBDO.

3.3. HSS-based MINO

The HSS is a meta-heuristic optimization algorithm inspired by a multidimensional space search mechanism [37]. The search process for the global optimal value is conducted by reasonably selecting the hyper-sphere centers (SCs) and the random generation of particles. The initial population number N_p , SC number N_{SC} , angle change probability P_α , the maximum and minimum radius of HSS $r_{\max}$ and $r_{\min}$ are determined in the particle initialization phase. The design variable vector $\mathbf{X}_i$ is randomly generated from $\mathbf{X}_d^L$ and $\mathbf{X}_d^U$ with a uniform probability. Notably, the continuous and integer design variables should be generated separately for the proposed MINO as:

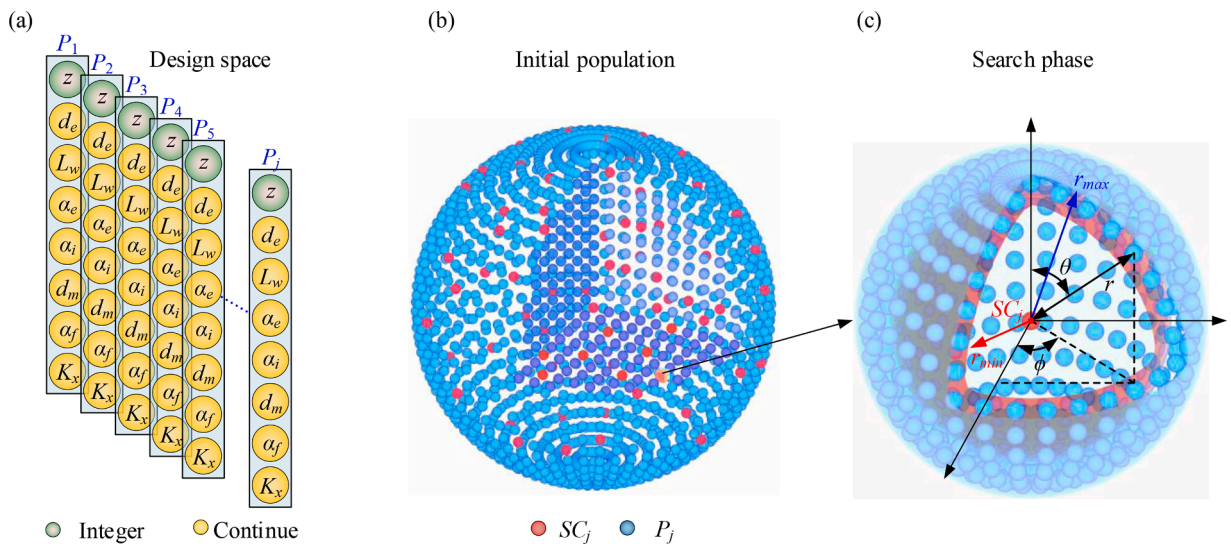


Fig. 6. Schematic diagram of MINO method based on HSS: (a) the generation strategy of the mixed-integer design variables, (b) the selection of the j^{th} hyper-sphere center SC_j and particles P_j , (c) the search procedure for the j^{th} hyper-sphere center SC_j .

$$\mathbf{T}_{init} = \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \vdots \\ \mathbf{P}_j \end{bmatrix} = [\mathbf{X}_{d1}, \mathbf{X}_{d2}, \dots, \mathbf{X}_{dn}, \mathbf{X}_{c1}, \mathbf{X}_{c2}, \dots, \mathbf{X}_{ck}] \begin{cases} \mathbf{X}_{dn} = randi, & randi \in (\mathbf{X}_{dn}^L, \mathbf{X}_{dn}^U) \\ \mathbf{X}_{ck} = \mathbf{X}_{ck}^L + rand(\mathbf{X}_{ck}^U - \mathbf{X}_{ck}^L) \end{cases}, \quad (34)$$

where $\mathbf{T}_{init}$ is the initial candidate set, $\mathbf{P}_j$ is the particle in the search space, $\mathbf{X}_{dn}$ is the n^{th} integer design variable, and $\mathbf{X}_{ck}$ is the k^{th} continuous design variable. For the proposed TRBDO, the generation of the $\mathbf{T}_{init}$ is shown in Fig. 6 (a). The initial solution set for each particle is generated by the proposed penalty function $f_R(\mathbf{P}_j)$ (Eq. (25)). Subsequently, the particles are sorted by the value of $f_R(\mathbf{P}_j)$. Meanwhile, the SCs are selected according to the best N_{sc} particles, as shown by the red particles in Fig. 6 (b).

The remaining particles are dominated by the objective function difference that can be expressed as:

$$D_i = \left| \frac{f_i - \max\{f_{sc}\}}{\sum_{i=1}^{N_{sc}} [f_i - \max\{f_{sc}\}]} \right| i = 1, 2, \dots, N_{sc}, \quad (35)$$

where f_{sc} is the SC set, and f_i is the penalty function value of the i^{th} SC in f_{sc} . The remaining particles of the i^{th} SC are randomly assigned by $D_i \times (N_p - N_{sc})$.

Spherical coordinate systems centered on SCs are established in the search phase. Then, the better particle within the bounded sphere is searched with the predefined SCs. In the 3-dimensional case, the position of the particle in search space can be changed by θ , ϕ , and r , as shown in Fig. 6 (c). Therefore, for the N -dimensional MINO, the position of the search particle can be defined by $N-1$ angles, which are randomly generated by $\alpha \in (0, 2\pi)$ with the probability of P_α . In addition, the radius is randomly selected in $(r \times r_{\min}, r \times r_{\max})$. The r for N -dimensional optimization can be expressed as:

$$r^2 = \sum_{i=1}^N (P_{i,sc} - P_{i,particle})^2, \quad (36)$$

where $P_{i,sc}$ is the SC in the i^{th} dimension and $P_{i,particle}$ is the particle around SC for the i^{th} dimension. The new SC is swapped by determining the lower penalty function values between the original SC and the search particle.

The particles searched in inappropriate spaces are determined as dummy particles. In this phase, the particle sets should be classified by the set objective function, which can be expressed as:

$$S_{of} = f_{sc} + \gamma \cdot \text{mean}\{f_{particle,sc}\}, \quad (37)$$

where f_{sc} and $f_{particle,sc}$ are the penalty function values of SC and corresponding particles, $\text{mean}\{f_{particle,sc}\}$ is the mean value of the penalty function value for all particles. It is worth noting that a low γ value will result in S_{of} being dominated by the penalty function value. Therefore, it is important to select an appropriate γ to increase the impact of particles on S_{of} . In this paper, γ is set as 0.1 [37]. These dummy particles should be reassigned to the correct SC to execute the appropriate search. Therefore, the difference of the particle sets can be defined as:

$$D_{sof} = S_{of} - \max\{S_{of}\}, \quad (38)$$

where $\max\{S_{of}\}$ is the maximum value among all particle sets. Subsequently, the dummy particles can be reassigned to other SCs by D_{sof} with the assigning probability P_{ra} , which can be expressed as:

$$P_{ra} = \left| \frac{NTOF_i}{\sum_{i=1}^{N_{sc}} NTOF_i} \right| i = 1, 2, \dots, N_{sc}, \quad (39)$$

where $NTOF$ is the normalized total objective function.

The aforementioned phases may obtain the particles with lower penalty function values than the corresponding SC. However, there also exist some worst particles with the highest penalty function values that are not expected to search for the optimal solution. To maintain the quality and quantity of the population, the $N_{new} = 0.05 \times N_p$ worst particles are swapped by the new particles (NPs) [38]. For the proposed MINO, the generation strategy of NPs can follow Eq. (34). These NPs are assigned to SC according to Eq. (35).

All particles and SCs are sorted based on their penalty function value. The best N_{sc} particles are selected as SCs for the next iteration. The optimization process is terminated by satisfying the maximum iteration number N_{max} or reaching the minimum iteration error (10^{-5}) compared to the previous $N_{break} = 50$ iterations. The flowchart of the proposed TRBDO for the main shaft bearing is displayed in Fig. 7.

4. Numerical example

For modern wind turbines, the design life of their main shaft bearings is generally demanded as 20 years [26]. The stochastic load and structure seriously impact the fatigue life and further reduce the service life of the main shaft bearing. In this section, the fatigue load spectrum is obtained based on a measured 10-minute stochastic loading history to characterize the randomness of the load, as shown in Fig. 3 (c). The proposed TRBDO method is applied to TRB, which is installed on the main shaft of a 5-MW off-shore wind turbine. The lubricant is Mobil 460WT with an operating temperature of 40 °C. The original bearing structure and lubricant performance are listed in Table 1. All numerical examples are tested on a personal computer with Intel i7-9700 and 16 GB of RAM.

4.1. Model validation

This subsection presents a comparative study and discusses the applicability of the bearing model to the TRBDO method. The bearing model in Appendix A, the Newton-Raphson method (NRM)-based model [39], and the FEM-based model [28] are employed. The original structural dimensions of the main shaft bearing are shown in Table 1.

Fig. 8 illustrates the internal contact characteristics of the TRB obtained through various modeling approaches. The results show that the non-load region increases with radial clearance. Simultaneously, the maximum contact load demonstrates a decreasing trend at first and then increases with the radial clearance. However, both the NRM-based and FEM-based bearing models may encounter non-convergence domains, which are primarily caused by the initial value selection and mesh scale. Notably, the equivalent dynamic load Q_e (Eq. (B.2)) will be determined as zero when the TRB model fails to converge, as shown in Figs. 8 (c)-(f). Meanwhile, the fatigue life will be assessed as infinite when Q_e is equivalent to zero. This scenario is not permitted during the TRBDO process.

The CPU time and average relative error (Eq. (A.6)) are separately recorded, as shown in Table 2. It can be seen that the model in Appendix A exhibits more prominent convergence and efficiency

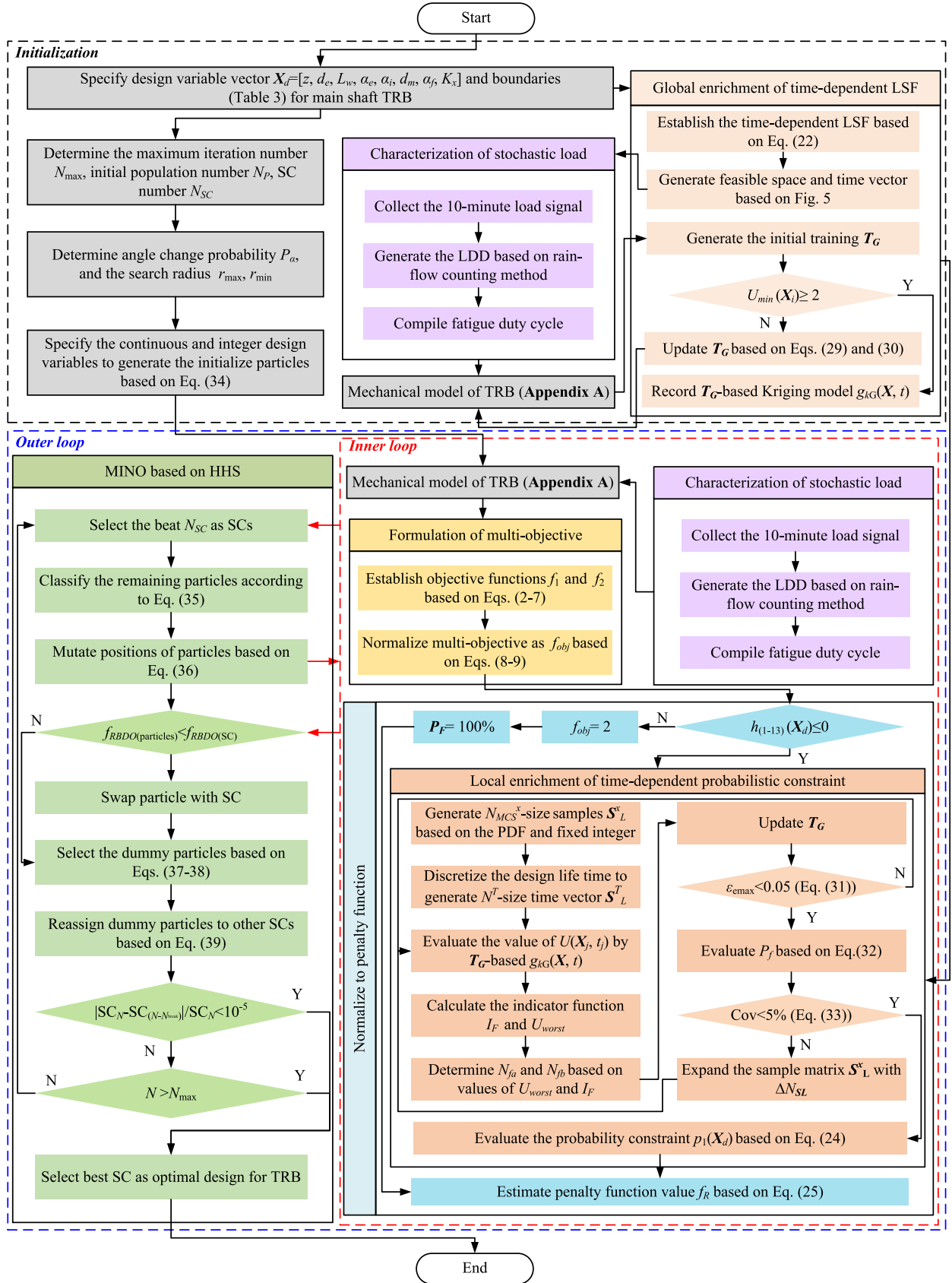


Fig. 7. Flowchart of the two-stage enrichment-based TRBDO approach for main shaft bearing.

Table 1

The original TRB structure and lubricant performance.

Parameter	Value	Parameter	Value
Outer diameter D (mm)	1580	Rear face thickness of the outer raceway O_r (mm)	63.5
Inner diameter d (mm)	1270	Width of the inner raceway B (mm)	200
Pitch diameter d_m (mm)	1425	Corner radius of the roller large-end c_1 (mm)	1.5
Number of rollers z	55	Corner radius of the roller small-end c_2 (mm)	1.2
Contact length of roller L_w (mm)	141	Minimum width of the flange l_f (mm)	35.9
Outer race contact angle α_e (°)	15.5	Rear flange diameter d_{il} (mm)	1425
Inner race contact angle α_i (°)	14	Front flange diameter d_{ir} (mm)	1336
Coefficient of profile K_x	0.00045	Modulus of elasticity E (Gpa)	210
Flange angle α_f (°)	75.8	Poisson's ratio ξ	0.3
Mean roller diameter d_e	73.23	Lubricant kinematic viscosity ν_{oil} (cst)	460
Width of the outer raceway C (mm)	152	Lubricant density ρ_{oil} (kg/m ³)	850

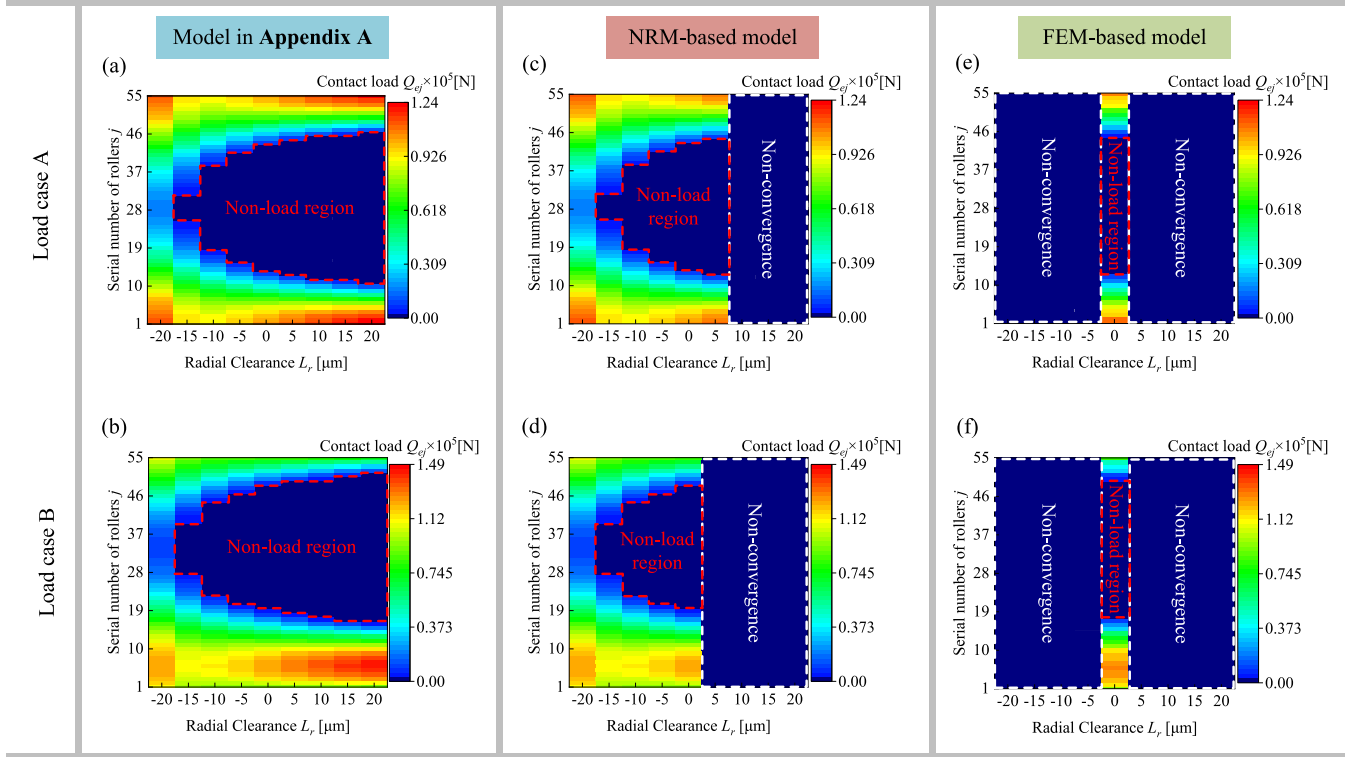


Fig. 8. Quantitative comparison of various TRB models under combined loads (Case A: $F_y = 1.2 \times 10^6$ N, $F_z = 0$ N, $M_y = 0$ N-mm, $M_z = 2.3 \times 10^8$ N-mm; Case B: $F_y = 1.2 \times 10^6$ N, $F_z = 8 \times 10^5$ N, $M_y = 1.5 \times 10^8$ N-mm, $M_z = 2.3 \times 10^8$ N-mm): the internal contact characteristic of TRB obtained by (a, b) the model in **Appendix A**, (c, d) the NRM-based model with initial value $[0, 0.05, 0, 0]$, and (e, f) FEM-based model.

capabilities than the NRM-based and FEM-based models. Although the FEM-based model exhibits higher fidelity compared to the applied model, its convergence and computational speed are significantly affected by the mesh scale and multiple contact regions. Notably, the

mechanical model of the TRB is integrated throughout the TRBDO process. Intuitively, the bearing model must be restructured for diverse stochastic configurations. The robustness of the bearing model directly relates to the implementation of TRBDO. According to the quantitative

Table 2

The comparison of the computational accuracy and efficiency among various models.

Radial clearance L_r	Case A Appendix A	NRM-based model	FEM-based model	Case B Appendix A	NRM-based model	FEM-based model
-20 μ m	0.0810 s (0)	0.0443 s (0.0172)	N/A (N/A)	0.1047 (0)	0.0675 (0.0072)	N/A (N/A)
-15 μ m	0.0804 s (0)	0.0577 s (0.0226)	N/A (N/A)	0.0852 (0)	0.1981 (0.0132)	N/A (N/A)
-10 μ m	0.0804 s (0)	0.2214s (0.0285)	N/A (N/A)	0.0744 (0)	0.4922 (0.0185)	N/A (N/A)
-5 μ m	0.0991 s (0)	0.4471 s (0.0192)	N/A (N/A)	0.1452 (0)	0.8434 (0.0112)	N/A (N/A)
0 μ m	0.0926 s (0)	0.5261 s (0.0144)	272.21 s (0.1624)	0.0941 (0)	1.5714 (0.0072)	297.80 s (0.1750)
5 μ m	0.0806 s (0)	1.1029s (0.0172)	N/A (N/A)	0.0745 (0)	N/A(N/A)	N/A (N/A)
10 μ m	0.0826 s (0)	N/A(N/A)	N/A (N/A)	0.0879 (0)	N/A(N/A)	N/A (N/A)
15 μ m	0.1293s (0)	N/A(N/A)	N/A (N/A)	0.0645 (0)	N/A(N/A)	N/A (N/A)
20 μ m	0.0833 s (0)	N/A(N/A)	N/A (N/A)	0.1254 (0)	N/A(N/A)	N/A (N/A)

Note: The numbering inside and outside parentheses indicate the average relative errors (Eq. (A.6)) and CPU times. N/A indicates not applicable.

comparison of convergence, accuracy, and efficiency, the bearing model in **Appendix A** is adopted in this paper to carry out TRBDO.

4.2. Application of TRBDO

This subsection demonstrates the practical engineering application of the proposed TRBDO with mixed-integer variables. It is assumed that the structure of the main shaft bearings is in the initial design phase. The statistical characteristics and type of variables can be selected based on the current machining accuracy, assembly capabilities, and the limit deviations of the design blueprint [28], as shown in **Table 3**. The target reliability R_{20}^T for the 20 years is determined as 97%.

For the proposed TRBDO, the initial population number N_p , SC number N_{SC} , and maximum iteration number N_{max} are determined as 100, 50, and 200. The reason for using high values of these control parameters is to obtain a sufficient candidate solution set. Additionally, inherent stochastic factors such as meta-heuristic optimization and failure probability assessment influence the optimal bearing structure obtained by the proposed TRBDO method. It is necessary to perform a statistical assessment for the optimization results. Thus, the proposed TRBDO is randomly executed 10 times in the same running environment, as shown in **Fig. 9**.

Fig. 9 (a) illustrates the convergence curves of 10 executions. It can be seen that the convergence steps of 10 executions are less than 200 iterations. The maximum termination steps among all tests occur during the 10th running operation, as shown by green bars in **Fig. 9** (b). Moreover, the minimum penalty function values of 10 times executions range from 1.458 to 1.477. The maximum relative error of the optimal penalty function value is less than 1.30%, which indicates the robustness of the proposed TRBDO method, as shown by the blue bars in **Fig. 9** (b).

Table 3

The statistical characteristics of variables for the proposed TRBDO.

Physical implication	Distribution type	Mean	Cov	Lower bound	Upper bound
Mean roller diameter d_e (mm)	normal	μ_{de}	0.015	50	90
Contact length of roller L_w (mm)	normal	μ_{Lw}	0.007	100	200
Outer race contact angle α_e (°)	normal	μ_{ae}	0.001	10	30
Inner race contact angle α_i (°)	normal	μ_{ai}	0.001	10	30
Pitch diameter d_m (mm)	normal	μ_{dm}	0.001	1396.5	1624.5
Flange angle α_f (°)	normal	μ_{af}	0.001	45	90
Coefficient of profile K_x	normal	μ_{Kx}	0.03	0.0001	0.001
Roller number z	–	μ_z	–	20	80
Radial clearance L_r (μm)	normal	20	0.5	–	–
Rotation speed N_e (rpm)	normal	11	0.03	–	–

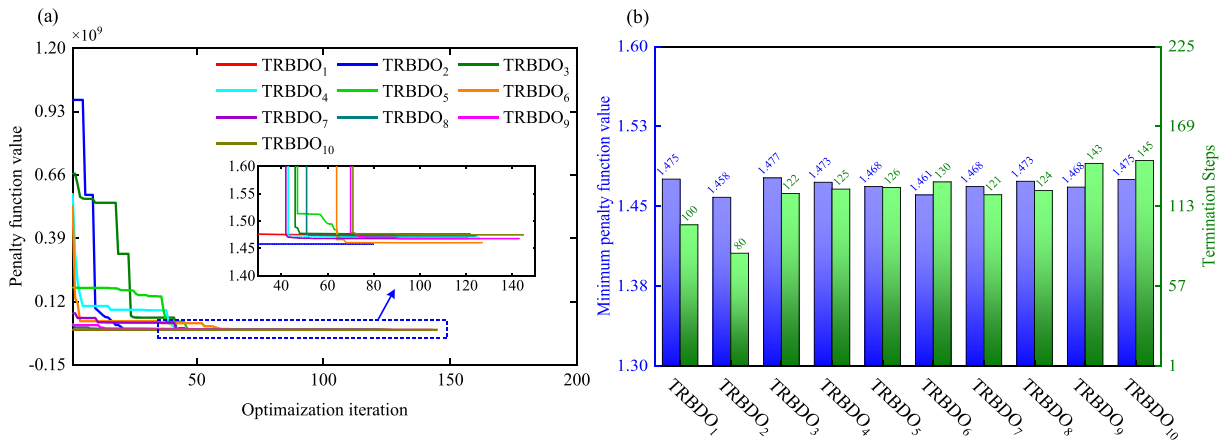


Fig. 9. The convergence test of the proposed TRBDO approach: (a) convergence curves of 10 executions, (b) quantitative comparison of termination steps and minimum penalty function values for 10 random executions.

4.3. Optimization capability of TRBDO

This subsection presents the geometric configuration and performance variations of the main shaft bearing before and after optimization. The optimal structure of the main shaft bearing is recorded and compared with the original design, as shown in **Fig. 10**. The results show that the proposed TRBDO searches for the high value of mean roller diameter, pitch diameter, flange angle, profile coefficient, and contact angles to obtain excellent comprehensive performance. Simultaneously, the corresponding roller number and contact length are decreased.

The two-stage enrichment approach in **Section 3** is adopted to evaluate the time-dependent reliability of original and optimized bearing structures. Three key performance indicators, fatigue life L , minimum oil film thickness h_{min}^0 , and time-dependent reliability P_F , are considered within the scope of the comparison, as shown in **Fig. 11**. Compared with the original structure, the optimal structure obtained by the proposed TRBDO expands the safe domains and further improves the reliability throughout the TRB service life, as shown by the blue line in **Fig. 11** (a). Moreover, the MCS approach is applied to verify the accuracy of the time-dependent failure probability of bearings before and after optimization, as shown by black and red balls in **Fig. 11** (a).

The comparison of the fatigue life L , minimum oil film h_{min}^0 , and fatigue failure probability P_F between the original and optimal structures is shown in **Fig. 11** (b). The results show that the fatigue life of the original TRB structure under the fatigue load spectrum is 117.14×10^6 revs, whereas the proposed TRBDO approach can improve the fatigue life of TRB by 41.1%, as shown by the blue bar in **Fig. 11** (b). Compared with the original structure, the proposed TRBDO approach can increase the minimum oil film thickness from 1.1805 μm to 1.2222 μm and further improve the wear resistance of the main shaft bearing, as shown

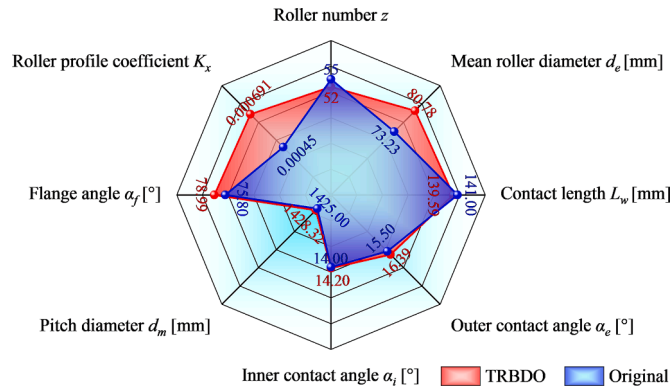


Fig. 10. The variation of mixed-integer design variables before and after TRBDO.

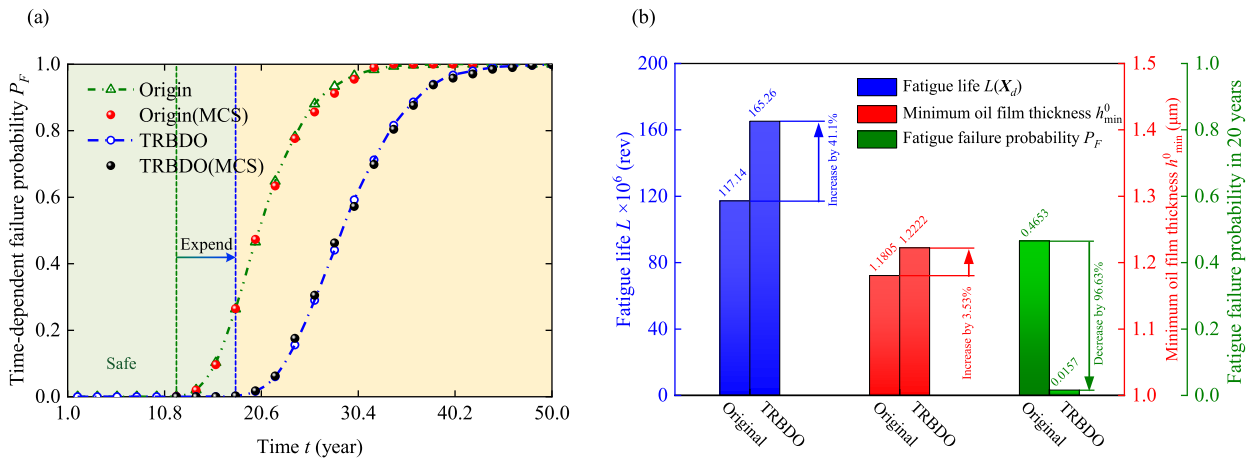


Fig. 11. Variations of the comprehensive performance of the main shaft bearing under the fatigue load spectrum.

by the red bar in Fig. 11 (b). In addition, the fatigue failure probability in 20 years is decreased by 96.63% based on the TRBDO, which indicates the effectiveness of the proposed method.

5. Conclusions

In this paper, a TRBDO approach for main shaft bearings in wind turbines was presented. The proposed approach handles the randomness of design, installation, and wind loads during the optimization to improve the fatigue resistance and lubrication condition of the main shaft bearing. A two-stage enrichment strategy was proposed to handle probability constraints with mixed-integer variables. In addition, the MINO method was developed for the formulated TRBDO problem based on HSS. The conclusions are detailed below.

Firstly, a mathematical model of TRBDO with mixed-integer variables was developed for main shaft bearings. A bearing model was applied to account for the randomness of combined loads, assembly quality, and geometries in the optimization procedure. The recommended bearing model exhibits superior efficiency and stability compared to NRM-based and FEM-based models. Therefore, the provided bearing model is applicable to TRBDO with mixed-integer variables.

Subsequently, a two-stage enrichment strategy was proposed to handle time-dependent probability constraints with integer design variables. Furthermore, the MINO method was developed based on a meta-heuristic algorithm to solve the formulated TRBDO problem. Ten random optimization results have shown that the maximum relative error and termination steps of the optimal penalty function value for 10

executions are less than 1.30% and 200, respectively, proving the robustness of the proposed TRBDO.

Eventually, the optimized results of the main shaft bearing have shown that the proposed TRBDO can increase the fatigue life and minimum oil film thickness by approximately 41.1% and 3.53%, respectively. Meanwhile, the fatigue failure probability in 20 years can decrease by approximately 96.63%. Hence, the proposed TRBDO approach can be adopted to design reliable bearing structures with adequate fatigue life and lubrication status for wind turbines.

CRediT authorship contribution statement

Zhiyuan Jiang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Xianzhen Huang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Bingxiang Wang:** Writing – review & editing, Supervision. **Xin Liao:** Writing – review & editing, Data curation. **Huizhen Liu:** Writing – review & editing, Supervision. **Pengfei Ding:** Writing – review & editing, Supervision.

Declaration of Competing Interest

There are no conflicts of interest to declare.

Data availability

No data was used for the research described in the article.

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Appendix A: Mechanical model of TRB

For the TRB subject to the combined load, the inner raceway is assumed to generate the displacement vector $U = [\delta_x, \delta_y, \theta_y, \theta_z]$, as shown in Fig. 1. The displacement of the j^{th} roller in the coordinate system $x_j y_j \theta$ can be derived as:

$$\begin{cases} \delta_{xj} = \delta_z \cos(\varphi_j - \varphi_{jr}) - \theta_y x_p \sin(\varphi_j) - \theta_z x_p \cos(\varphi_j) - L_r/2 \\ \delta_{yj} = \delta_x + (\theta_y d_m \sin(\varphi_j) + \theta_z d_m \cos(\varphi_j))/2 - L_a \\ \varphi_{\theta} = \varphi_j \sin(\varphi_j) + \varphi_z \cos(\varphi_j) \\ \varphi_{jr} = \arctan(F_z/F_y) \end{cases}, \quad (\text{A.1})$$

where $\varphi_j = 2\pi(j-1)/z$ is the azimuth angle of the j^{th} roller, φ_{jr} is the conversion angle determined by the combined load, L_r and $L_a = L_r/2 \tan(\alpha_e)$ are the radial and axial clearance. Furthermore, the displacement of the j^{th} roller in the coordinate system $x_{gj} y_{gj} \psi$ can be expressed as:

$$\begin{cases} \delta_{y_{gj}} = \delta_{yj} \cos(\alpha_m) + \delta_{xj} \sin(\alpha_m) \\ \delta_{x_{gj}} = -\delta_{yj} \sin(\alpha_m) + \delta_{xj} \cos(\alpha_m), \\ \varphi_{\psi} = \varphi_{\theta} \end{cases}, \quad (\text{A.2})$$

where $\alpha_m = (\alpha_e + \alpha_i)/2$ is the mean contact angle. Based on the slicing technology, the normal displacement δ_{ejk} of the outer raceway for the k^{th} slice can be expressed as:

$$\delta_{ejk} = \delta_{y_{gj}} \cos(\varepsilon) + \delta_{x_{gj}} \sin(\varepsilon) - h(l_k) - l_k \varphi_{\psi}, \quad (\text{A.3})$$

where $\varepsilon = (\alpha_e - \alpha_i)/2$ is the Semi-taper angle, $h(\cdot)$ is the roller profile function, and l_k is the position of the k^{th} slice in the coordinate system $h x_s$, as shown in Fig. 2 (b).

The contact load of the k^{th} slice can be expressed as:

$$\begin{cases} q_{ejk} = K_l \delta_{ejk}^{10/9} > 0 \\ q_{ejk} = 0 & \delta_{ejk} \leq 0 \end{cases}, \quad (\text{A.4})$$

where $K_l = 8.06 \times 10^4 (\Delta l)^{8/9}$ is the contact stiffness between the roller slice and the raceway. The balance equations of the inner raceway can be expressed as:

$$\begin{cases} \sum_{j=1}^z \left[\sum_{k=1}^n q_{ejk} \sin(\alpha_e) \right] - F_x = 0 \\ \sum_{j=1}^z \left\{ \left[\left(\sum_{k=1}^n q_{ejk} \right) \cos(\alpha_e) \right] \cos(\varphi_j - \varphi_{jr}) \right\} - \sqrt{F_y^2 + F_z^2} = 0 \\ \sum_{j=1}^z \left\{ \sum_{k=1}^n q_{ejk} [d_m/2 - x_p/\tan(\alpha_e)] \sin(\alpha_e) - l_k \right\} \sin(\varphi_j) - M_y = 0 \\ \sum_{j=1}^z \left\{ \sum_{k=1}^n q_{ejk} [d_m/2 - x_p/\tan(\alpha_e)] \sin(\alpha_e) - l_k \right\} \cos(\varphi_j) - M_z = 0 \end{cases}. \quad (\text{A.5})$$

The displacement vector U can be determined by solving nonlinear equations Eq. (A.5). This paper adopted the strategy of meta-heuristic optimization to solve Eq. (A.5). The details of the solution can be referred to [26].

The average relative error between the proposed model and other models can be expressed as follows:

$$\varepsilon_A = \frac{\left| \sum_{j=1}^z [(Q_{pej} - Q_{nej})/Q_{nej}] \right|}{z}, \quad (\text{A.6})$$

where Q_{pej} and Q_{nej} are contact loads obtained through the proposed model and the comparative model, respectively.

Appendix B: Fatigue life of TRB

According to the standards of ISO 281 and ISO 16,281, the basic dynamic capacities of the inner and outer raceway can be expressed as [28]:

$$\begin{cases} Q_{ci} = 552b_m\lambda \frac{(1-\gamma)^{29/27}}{(1+\gamma)^{1/4}} \cos^{-2/9}(\alpha_m) d_e^{29/27} L_w^{7/9} z^{-1/4} \gamma^{2/9} \\ Q_{ce} = 552b_m\lambda \frac{(1+\gamma)^{29/27}}{(1-\gamma)^{1/4}} \cos^{-2/9}(\alpha_m) d_e^{29/27} L_w^{7/9} z^{-1/4} \gamma^{2/9} \end{cases}, \quad (\text{B.1})$$

where $b_m = 1.1$ is the modified coefficient for TRB, $\lambda=0.6$ is the reduction coefficient, and $\gamma = d_e \times \cos(\alpha_m)/d_m$ is the scale factor of the roller thickness.

The equivalent dynamic loads of the nonrotating and rotating raceways can be expressed as:

$$\begin{cases} Q_{ei} = \left[\left(\sum_{j=1}^z Q_{ij}^{9/2} \right) / z \right]^{2/9} \\ Q_{eo} = \left[\left(\sum_{j=1}^z Q_{ej}^4 \right) / z \right]^{1/4} \end{cases}, \quad (\text{B.2})$$

where Q_{ij} is the contact load between the j^{th} roller and the inner raceway, and Q_{ej} is the contact load between the j^{th} roller and the outer raceway. The values of Q_{ij} and Q_{ej} for each roller can be obtained by the proposed TRB mechanical model, as shown in Fig. 1 and Appendix A.

Appendix C: Kriging surrogate model

The Kriging surrogate model $g_k(x)$ can be expressed as [40]:

$$g_k(X) = \sum_{i=1}^P f^T(X) \beta + z(X), \quad (\text{C.1})$$

where $f(X) = [f_1(X), f_2(X), \dots, f_p(X)]$ is the regression function vector of X , $\beta = [\beta_1, \beta_2, \dots, \beta_p]^T$ is the corresponding regression coefficient vector for f , and $z(X)$ is a stationary Gaussian process with the mean value of zero and a standard deviation σ . The covariance function of $z(X)$ can be obtained as:

$$\text{Cov}[z(X^{(i)}), z(X^{(j)})] = \sigma^2 R(X^{(i)}, X^{(j)}, \theta), \quad (\text{C.2})$$

where $R(X^{(i)}, X^{(j)}, \theta)$ is the auto-correlation function, which can be expressed as:

$$R(X^{(i)}, X^{(j)}, \theta) = \exp \left(- \sum_{k=1}^m \theta_k \left| X_k^{(i)} - X_k^{(j)} \right|^2 \right), \quad (\text{C.3})$$

where X_k is the k^{th} dimension element of $X^{(i)}$, $\theta = [\theta_1, \theta_2, \theta_3, \dots, \theta_k]^T$ is the correlation hyperparameters vector for R , and it can be evaluated by the maximum likelihood estimation:

$$\max_{\theta} F(\theta) = - \frac{m \ln(\hat{\sigma}^2) + \ln(R)}{2} \theta_k \geq 0 (k = 1, 2, \dots, m), \quad (\text{C.4})$$

where R is correlation matrix composed of R , $\hat{\sigma}^2$ is predicted value of σ^2 , which can be obtained by:

$$\hat{\sigma}^2 = \frac{(g - F\hat{\beta})^T R^{-1} (g - F\hat{\beta})}{m}, \quad (\text{C.5})$$

where g is the response for X , F is the matrix composed of regression model for m -dimensional sample points, $\hat{\beta}$ is the predicted value of β , and it can be calculated by:

$$\hat{\beta} = (F^T R^{-1} F)^{-1} F^T R^{-1} g. \quad (\text{C.6})$$

The predicted value $\mu_{gk}(X)$ and corresponding variance $\sigma_{gk}(X)$ of the Kriging surrogate model can be obtained by:

$$\begin{cases} \mu_{gk}(X) = f^T(X) \hat{\beta} + r_c^T(X) R^{-1} (g - F\hat{\beta}) \\ \sigma_{gk}^2(X) = \hat{\sigma}^2 \left\{ 1 - r_c^T(X) R^{-1} r_c(X) + [F^T R^{-1} r_c(X) - f(X)]^T (F^T R^{-1} F)^{-1} [F^T R^{-1} r_c(X) - f(X)] \right\} \end{cases}. \quad (\text{C.7})$$

References

- [1] Papi F, Bianchini A. Technical challenges in floating offshore wind turbine upscaling: a critical analysis based on the NREL 5 MW and IEA 15 MW Reference Turbines. *Renew Sustain Energy Rev* 2022;162:112489.
- [2] Council G W E. GWEC global wind report 2019. Bonn, Germany: Global Wind Energy Council; 2017.
- [3] Kaewniam P, Cao M, Alkayem N F, et al. Recent advances in damage detection of wind turbine blades: a state-of-the-art review. *Renew Sustain Energy Rev* 2022;167:112723.
- [4] Negm H M, Maalawi KY. Structural design optimization of wind turbine towers. *Comput Struct* 2000;74(6):649–66.
- [5] Li Y, Wei K, Yang W, et al. Improving wind turbine blade based on multi-objective particle swarm optimization. *Renew Energy* 2020;161:525–42.

- [6] Hegseth J M, Bachynski E E, Martins J R R A. Integrated design optimization of spar floating wind turbines. *Mar Struct* 2020;72:102771.
- [7] Zhang C, Long K, Zhang J, et al. A topology optimization methodology for the offshore wind turbine jacket structure in the concept phase. *Ocean Eng* 2022;266: 112974.
- [8] Lu F, Long K, Zhang C, et al. A novel design of the offshore wind turbine tripod structure using topology optimization methodology. *Ocean Eng* 2023;280:114607.
- [9] Ling C, Kuo W, Xie M. An overview of adaptive-surrogate-model-assisted methods for reliability-based design optimization. *IEEE Trans Reliab* 2022.
- [10] Chaudhuri A, Kramer B, Willcox KE. Information reuse for importance sampling in reliability-based design optimization. *Reliab Eng Syst Saf* 2020;201:106853.
- [11] Liang J, Mourelatos Z P, Nikolaidis E. A single-loop approach for system reliability-based design optimization. *J Mech Des* 2007;129(12):1215–24.
- [12] Zhang X, Lu Z, Cheng K. Reliability index function approximation based on adaptive double-loop Kriging for reliability-based design optimization. *Reliab Eng Syst Saf* 2021;216:108020.
- [13] Kuschel N, Rackwitz R. Two basic problems in reliability-based structural optimization. *Math Method Oper Res* 1997;46:309–33.
- [14] Ma Y Z, Jin X X, Wu X L, et al. Reliability-based design optimization using adaptive Kriging-A single-loop strategy and a double-loop one. *Reliab Eng Syst Saf* 2023; 237:109386.
- [15] Moustapha M, Sudret B. Surrogate-assisted reliability-based design optimization: a survey and a unified modular framework. *Struct Multidiscip Optim* 2019;60: 2157–76.
- [16] Thapa M, Missoum S. Surrogate-based stochastic optimization of horizontal-axis wind turbine composite blades. *Struct Multidiscip Optim* 2022;65(2):41.
- [17] Okoro A, Khan F, Ahmed S. Dependency effect on the reliability-based design optimization of complex offshore structure. *Reliab Eng Syst Saf* 2023;231:109026.
- [18] Leimeister M, Kolios A. Reliability-based design optimization of a spar-type floating offshore wind turbine support structure. *Reliab Eng Syst Saf* 2021;213: 107666.
- [19] Zhao Z, Zhao Y G, Li P P. A novel decoupled time-variant reliability-based design optimization approach by improved extreme value moment method. *Reliab Eng Syst Saf* 2023;229:108825.
- [20] Li J, Chen J. Solving time-variant reliability-based design optimization by PSO-t-IRS: a methodology incorporating a particle swarm optimization algorithm and an enhanced instantaneous response surface. *Reliab Eng Syst Saf* 2019;191:106580.
- [21] Liu G, Liu H, Zhu C, et al. Design optimization of a wind turbine gear transmission based on fatigue reliability sensitivity. *Front Mech Eng* 2021;16:61–79.
- [22] Verma S K, Tiwari R. Robust optimum design of tapered roller bearings based on maximization of fatigue life using evolutionary algorithm. *Mech Mach Theory* 2020;152:103894.
- [23] Baklouti A, Dammak K, El Hami A. Optimum reliable design of rolling element bearings using multi-objective optimization based on C-NSGA-II. *Reliab Eng Syst Saf* 2022;223:108508.
- [24] Cao H, Niu L, Xi S, et al. Mechanical model development of rolling bearing-rotor systems: a review. *Mech Syst Sig Process* 2018;102:37–58.
- [25] Habibi H, Howard I, Simani S. Reliability improvement of wind turbine power generation using model-based fault detection and fault tolerant control: a review. *Renew Energy* 2019;135:877–96.
- [26] Jiang Z, Huang X, Zhu H, et al. A new method for contact characteristic analysis of the tapered roller bearing in wind turbine main shaft. *Eng Fail Anal* 2022;141: 106729.
- [27] Adedipe T, Shafiee M, Zio E. Bayesian network modelling for the wind energy industry: an overview. *Reliab Eng Syst Saf* 2020;202:107053.
- [28] Jiang Z, Huang X, Liu H, et al. Dynamic reliability analysis of main shaft bearings in wind turbines. *Int J Mech Sci* 2022;235:107721.
- [29] Palmgren A. Die lebensdauer von kugellagern. *Verfahrenstechnik* 1924;68: 339–41.
- [30] Liu Y, Chen Z, Li Y, et al. Dynamic investigation and alleviative measures for the skidding phenomenon of lubricated rolling bearing under light load. *Mech Syst Sig Process* 2023;184:109685.
- [31] Li H, Zhang Z, Li L, et al. State variable and optimization potential-based multi-objective optimization method and application in compressor blade airfoil design. *Struct Multidiscip Optim* 2023;66(7):165.
- [32] Jiang C, Yan Y, Wang D, et al. Global and local Kriging limit state approximation for time-dependent reliability-based design optimization through wrong-classification probability. *Reliab Eng Syst Saf* 2021;208:107431.
- [33] Song Z, Zhang H, Liu Z, et al. A two-stage Kriging estimation variance reduction method for efficient time-variant reliability-based design optimization. *Reliab Eng Syst Saf* 2023;237:109339.
- [34] Echard B, Gayton N, Lemaire M. AK-MCS: an active learning reliability method combining Kriging and Monte Carlo simulation. *Struct Saf* 2011;33(2):145–54.
- [35] Hu Z, Mahadevan S. A single-loop kriging surrogate modeling for time-dependent reliability analysis. *J Mech Des* 2016;138(6):061406.
- [36] Wang Z, Shafieezadeh A. ESC: an efficient error-based stopping criterion for kriging-based reliability analysis methods. *Struct Multidiscip Optim* 2019;59: 1621–37.
- [37] Rahmatian M, Seyedtabaai S. Multi-machine optimal power system stabilizers design based on system stability and nonlinearity indices using Hyper-Spherical Search method. *Int J Electr Power Energy Syst* 2019;105:729–40.
- [38] Sanjari M J, Karami H, Yatim A H, et al. Application of Hyper-Spherical Search algorithm for optimal energy resources dispatch in residential microgrids. *Appl Soft Comput* 2015;37:15–23.
- [39] Zheng J, Ji J, Yin S, et al. Internal loads and contact pressure distributions on the main shaft bearing in a modern gearless wind turbine. *Tribol Int* 2020;141:105960.
- [40] Zhao Z, Lu Z H, Zhao Y G. An efficient extreme value moment method combining adaptive Kriging model for time-variant imprecise reliability analysis. *Mech Syst Sig Process* 2022;171:108905.